

POLAR CO-ORDINATES

2

Art. 8. Polar Co-ordinates : In using the cartesian co-ordinates x, y , we indicate the position of a point in a plane in terms of its distances from two lines. It is sometimes preferable to show its location in terms of its distance from a fixed point and its direction from a fixed line through this point. The co-ordinates of points in this system are called **Polar co-ordinates**.

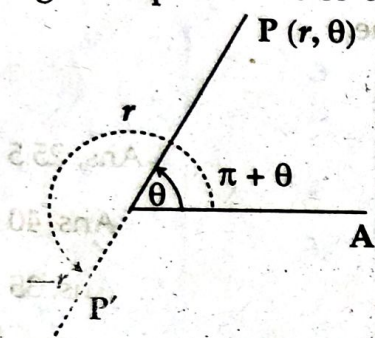


Fig:5

Let O be a fixed point and OA a fixed straight line. Let P be any point. Then the length of OP usually denoted by r , and the measure of the $\angle AOP$, usually denoted by θ are the polar co-ordinates of P.

O is called the Pole, or the origin and OA is called the initial line or the directed line OA is called the **Polar axis**. r is called the radius vector and θ is called the vectorial angle.

Thus the polar co-ordinates of P are expressed as (r, θ) . The vectorial angle is said to be positive when measured in the anticlockwise direction and negative in the clockwise direction. To find the position of P, first measure the vectorial angles θ and then cut off the radius vector equal to r .

In polar co-ordinates the same point is represented by an indefinitely large number of sets of co-ordinates. Thus

$$(r, \theta), (r, \theta + 2\pi), (r, \theta + 4\pi) \dots \dots \dots (r, \theta - 2\pi) - (r, \theta - 4\pi)$$

$$(-r, \theta + \pi); (r, \theta + 3\pi), \dots \dots \dots \text{and } (-r, \theta - \pi), (-r, \theta - 3\pi)$$

all represent the same point, viz, P, as the student will easily see on trying to plot the above co-ordinates.

Art. 9. Relation between the polar and cartesian co-ordinates.

Let the pole and the initial line of the polar system be the same as the origin and the positive direction of the x -axis of the cartesian system. Suppose the co-ordinates of a point be (r, θ) and cartesian co-ordinates (x, y)

Let the axes be rectangular

It is evident from fig (6) that

$$x = r \cos \theta, y = r \sin \theta$$

$$\text{Also, } x^2 + y^2 = r^2.$$

$$\frac{y}{x} = \tan \theta$$

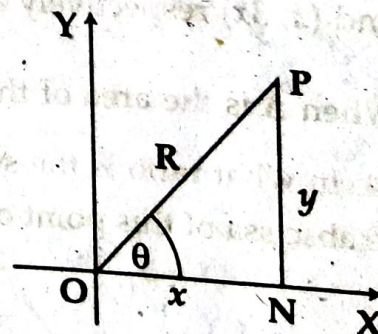


Fig:6

Thess formula are used for transformations from one system to the other system.

Art. 10. Distance between two points :

To find the distance between two points whose polar co-ordinates are given.

Let P_1 be the point (r_1, θ_1)

and P_2 the point (r_2, θ_2)

$$\angle POP_2 = \angle XOP_2 - \angle XOP_1 = \theta_2 - \theta_1$$

$$\text{Then } P_1P_2^2 = OP_1^2 + OP_2^2 - 2OP_1 OP_2 \cos P_1 OP_2$$

$$= r_1^2 + r_2^2 - 2r_1r_2 \cos (\theta_2 - \theta_1)$$

$$P_1P_2 = \sqrt{\{r_1^2 + r_2^2 - 2r_1r_2 \cos (\theta_2 - \theta_1)\}}$$

Art. 11. Area of a triangle :

Let P_1, P_2, P_3 be the three vertices whose polar co-ordinates are $(r_1, \theta_1), (r_2, \theta_2), (r_3, \theta_3)$ respectively. The area of the triangle $P_1P_2P_3$.

$$\Delta P_1P_2P_3 = \Delta OP_1P_2 + \Delta OP_2P_3 - \Delta OP_1P_3$$

$$= \frac{1}{2} r_1 r_2 \sin (\theta_2 - \theta_1) + \frac{1}{2} r_2 r_3 \sin (\theta_3 - \theta_2) - \frac{1}{2} r_1 r_3 \sin (\theta_3 - \theta_1)$$

$$= \frac{1}{2} r_1 r_2 \sin (\theta_2 - \theta_1) + \frac{1}{2} r_2 r_3 \sin (\theta_3 - \theta_2) - \frac{1}{2} r_3 r_1 \sin (\theta_1 - \theta_3)$$

Continuing on the same way, the area of a polygon

$P_1P_2P_3 \dots P_n$

$$= \frac{1}{2} \{ r_1 r_2 \sin (\theta_2 - \theta_1) +$$

$$r_2 r_3 \sin (\theta_3 - \theta_2) + \dots$$

$$\dots + r_{n-1} r_n \sin (\theta_n - \theta_{n-1}) + r_n r_1 \sin (\theta_1 - \theta_n) \}$$

Ex. 1. Find the polar co-ordinates of the point whose cartesian co-ordinates are

(i) $(-\sqrt{3}, 1)$, (ii) $(5, 12)$

$$(i) r = \sqrt{(3+1)} = 2, \tan \theta = -1/\sqrt{3} \therefore \theta = 5\pi/6$$

The polar co-ordinates are $(2, 5\pi/6)$

$$(ii) r = \sqrt{(25+144)} = 13, \tan \theta = 12/5, \theta = \tan^{-1} 12/5$$

The polar co-ordinates are $(13, \tan^{-1} 12/5)$

Ex. 2. Find the cartesian co-ordinates of the point whose polar co-ordinates are

(i) $(5, -\pi/4)$, (ii) $(2, 330^\circ)$

$$(i) x = 5 \cos (-\pi/4) = 5/\sqrt{2} = 5\sqrt{2}/2 \text{ and } y = 5 \sin (-\pi/4) = -5\sqrt{2}/2$$

The cartesian co-ordinates are $(5\sqrt{2}/2, -5\sqrt{2}/2)$

$$(ii) x = 2 \cos 330^\circ = 2 \cos (560^\circ - 30^\circ) = 2 \cos 30^\circ = \sqrt{3}, y = 2 \sin 330^\circ = -2 \sin 30^\circ = -1$$

The cartesian co-ordinates are $(\sqrt{3}, -1)$

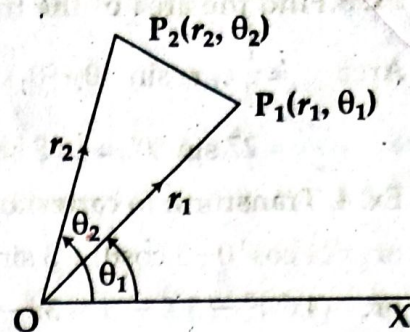


Fig. 7

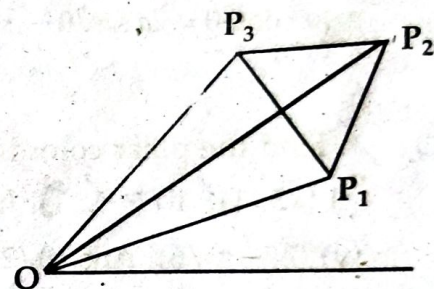


Fig. 7(1)

Ex. 3. Find the area of the triangle whose vertices are $(0,0)$, $(9, 20^\circ)$, and $(6, 50^\circ)$

$$\text{Area} = \frac{1}{2} r_1 r_3 \sin (\theta_2 - \theta_1) = \frac{1}{2} 6 \cdot 9 \sin (50^\circ - 20^\circ)$$

$$= 27 \sin 30^\circ = 13.5 \text{ square units.}$$

Ex. 4. Transform to cartesian co-ordinates the equation $r (\cos 3\theta + \sin 3\theta) = 5k \sin \theta \cos \theta$

$$\text{or, } r (4 \cos^3 \theta - 3 \cos \theta) + 3 \sin \theta - 4 \sin^3 \theta = 5k \sin \theta \cos \theta$$

$$\text{or, } r (4x^3/r^3 - 3x/r + 3y/r - 4y^3/r^3) = 5k x/r y/r + 4y/r$$

$$\text{or, } 4(x^3 - y^3) - 3(x^2 + y^2)(x - y) = 5kxy$$

Ex. 5. Transform to polar co-ordinates the equations

(i) $9x^2 + 4y^2 = 36$ (ii) $x^3 = y^2 (2a - x)$

(i) Putting $x = r \cos \theta$, $y = r \sin \theta$ the required equation is $9r^2 \cos^2 \theta + 4r^2 \sin^2 \theta = 36$ or, $r^2 (4 + 5 \cos^2 \theta) = 36$

(ii) $r^3 \cos^3 \theta = r^2 \sin^2 \theta (2a - r \cos \theta)$

or, $r^2 \cos^3 \theta = 2a \sin^2 \theta - r \sin^2 \theta \cos \theta$ or, $r \cos \theta = 2a \sin^2 \theta$

EXERCISE II

1. Find the polar co-ordinates of the points whose cartesian co-ordinates are

(i) $(\sqrt{3}, 1)$, (ii) $(-1, 1)$, (iii) $(\sqrt{3}, -1)$ (iv) $(5, \pi^2)$

(v) $(2, -\pi/6)$, Ans (i) $(2, \pi/6)$ (ii) $(\sqrt{2}, 3\pi/4)$

2. Find the polar distance between the points whose polar coordinates are

(i) $(-3, 45^\circ)$ and $(7, 105^\circ)$

Ans (79)

(ii) $(\sqrt{2}, 5\pi/4)$ and $(2, 2\pi/3)$

Ans $1 + \sqrt{3}$

3. Determine the nature of the triangle whose vertices are the points $(0, 0)$, $(3, \pi - 2)$ and $(3, \pi - 6)$ and calculate its area.

Ans. $+9(\sqrt{3}) - 4$.

4. Find the areas of the triangle the polar co-ordinates of whose angular points are

(i) $(-3, 30^\circ)$, $(5, 150^\circ)$ and $(7, 210^\circ)$

(ii) (a, θ) , $(2a, \theta + \pi/3)$, $(3a, \theta + 2\pi/3)$

Ans. $7(\sqrt{3})/2$

Ans. $(5\sqrt{3}/4)a^2$

5. Change the equations to polar Co-ordinates

(i) $(x^2 + y^2)^2 = 2a^2 xy$

(ii) $x^4 + x^2 y^2 - (x + y)^2 = 0$

Ans. $r^2 = a^2 \sin 2\theta$

Ans. $r = \pm (1 + \tan \theta)$

6. Transform to co-ordinates the equations

(i) $r^2 - 2r(\cos \theta - \sin \theta) - 7 = 0$, Ans. $x^2 + y^2 - 2x + 2y - 7 = 0$

(ii) $r(1 - e \cos \theta) = ep$ Ans. $x^2 + y^2 = e^2(p + x)^2$

7. A line of length $2a$ has its extremities on two fixed perpendicular lines. Find the locus of the foot of the perpendicular from the intersection of the two fixed lines to a given line.

Ans. $r = a \sin 2\theta$